Filter Trait Outliers Using IQR

Description

Removes outlier trait values from a long-format trait data frame using the interquartile range (IQR) method. Outlier detection is performed per group (by default per taxon x trait combination), mirroring the fence logic used by boxplots: \[Q1 - k*IQR, Q3 + k*IQR\].

Usage

```
filter_trait_outliers(

  data_trait_records,

  trait_value_column = "trait_value",

  grouping_columns = c("taxon_name", "trait_domain_name"),

  iqr_multiplier = 1.5,

  verbose = TRUE

)
```

Arguments

`data_trait_records`	A data frame in long format with at least the columns specified by 'trait_value_column' and 'grouping_columns'.
`trait_value_column`	A character string naming the numeric column containing trait values. Default: '"trait_value"'.
`grouping_columns`	A character vector of column names used to define groups for outlier detection. Default: 'c("taxon_name", "trait_domain_name")'.
`iqr_multiplier`	A positive numeric scalar controlling the fence width, equivalent to k in the standard boxplot formula. Default: '1.5'.
`verbose`	Logical scalar. If 'TRUE', print the number of removed records.

Details

Groups with an IQR of zero (all values identical) are kept intact — no filtering is applied to constant groups. This prevents inadvertent removal of valid data when all observations share the same trait value.

Value

A data frame with the same columns as the input but with outlier rows removed. The number of removed rows is reported via 'cli::cli_inform()'.

See Also

[flag_trait_outliers()], [aggregate_trait_values()]